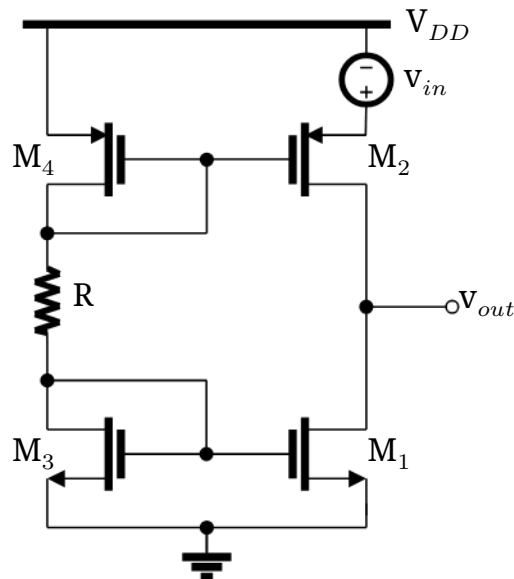


For this lab, use N_18_MM and P_18_MM MOSFETs from UMC180.

Task 1 : Design a Common Gate Amplifier (CGA) with PMOS as input device and NMOS as constant current source load as shown in diagram for $g_{m_2}=0.77 \text{ mS}$, $V_{DD}=1.8 \text{ V}$. The overdrive voltages of both NMOS and PMOS are same. $R=10 \text{ k}\Omega$. Calculate size of NMOS, PMOS, bias current in M_2 and gain of amplifier. M_1 and M_3 are same in size and M_2 and M_4 are same in size.

Task 2 : If R is replaced by diode connected NMOS without changing bias current, what should be the size of NMOS?



- Perform DC simulation to confirm that all NMOS and PMOS are in saturation in task 1 and task 2.
- Perform AC simulation to find out gain and bandwidth of your amplifier in task 1 and task 2.
- Perform transient simulation to plot input and output transient signals in task1 and task 2.

Use $\mu_n C_{ox}=300 \mu\text{A}/\text{V}^2$, $\lambda_n=0.11 \text{ V}^{-1}$ for $L=1 \mu\text{m}$, $V_{th0,n}=0.38 \text{ V}$.

$\mu_p C_{ox}=60 \mu\text{A}/\text{V}^2$, $\lambda_p=0.03 \text{ V}^{-1}$ for $L=1 \mu\text{m}$, $|V_{th0,p}|=0.5 \text{ V}$.